## TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

**Reminders**
**Words:** at least ≥ · at most ≤ · more than > · fewer than <.
**Solve like an equation:** undo + and −, then × and ÷.
**Flip the sign** when you multiply or divide both sides by a negative number.
**Graph:** ○ for < and >, ● for ≤ and ≥. Shade right for > and ≥, left for < and ≤.

### 1. What is the solution set for $-3x + 4 \geq 19$?

**Tip:** Dividing by a negative flips the sign.

**A.** $x \geq -5$

**B.** $x \leq 5$

**C.** $\boldsymbol{x \leq -5}$

$-3x \geq 15 \rightarrow$ divide by $-3$, flip $\rightarrow x \leq -5$

**2.** Solve the inequality, and graph the solution on the number line.
$5 - 2x < 11$

> **Tip:** Subtract 5, then divide by −2 and flip.

Subtract 5: −2x < _____

Divide by −2 and flip: x _____ _____

**$x > -3$, open circle at −3, shaded right**

$-2x < 6 \rightarrow x > -3$ (flip)

**3.** A tire has 26 psi and loses 0.1 psi per minute. It must stay at or above 22 psi. Write and solve an inequality for the time $t$ in minutes.

> **Tip:** At or above → ≥. Divide by −0.1 and flip.

26 − 0.1t _____ 22

−0.1t ≥ _____

t _____ _____

**$t \leq 40$ minutes**

$26 - 0.1t \geq 22 \rightarrow -0.1t \geq -4 \rightarrow t \leq 40$